

Using the vector product

Basic skills

Q1. a) $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times -1 - 0 \times 2 \\ 0 \times 5 - -2 \times -1 \\ -2 \times 2 - 1 \times 5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -9 \end{pmatrix}$

could use

$$\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -2 & 1 & 0 \\ 5 & 2 & -1 \end{vmatrix} = \underline{i} \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} - \underline{j} \begin{vmatrix} -2 & 0 \\ 5 & -1 \end{vmatrix} + \underline{k} \begin{vmatrix} -2 & 1 \\ 5 & 2 \end{vmatrix}$$
$$= -\underline{i} - 2\underline{j} - 9\underline{k}$$

b) $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

mag is pythagoras

Q2. $\sqrt{5^2 + 3^2 + 8^2} = \sqrt{98} = 7\sqrt{2}$

Q3. $\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 0 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \times 7 - 1 \times 0 \\ 1 \times -2 - 5 \times 7 \\ 5 \times 0 - 3 \times -2 \end{pmatrix} = \begin{pmatrix} 21 \\ -37 \\ 6 \end{pmatrix}$

magnitude of normal $\sqrt{21^2 + 37^2 + 6^2} = \sqrt{1846}$

so $\frac{1}{\sqrt{1846}} \begin{pmatrix} 21 \\ -37 \\ 6 \end{pmatrix}$ or $-\frac{1}{\sqrt{1846}} \begin{pmatrix} 21 \\ -37 \\ 6 \end{pmatrix}$ ← opposite direction

Q4. As $\underline{a} \times \underline{b} = |\underline{a}| |\underline{b}| \sin \theta \hat{n}$

If Parallel $\theta = 0$ so $\sin \theta = 0$ so

$$\underline{a} \times \underline{b} = \underline{0}$$

Q5. $\underline{i} \times \underline{j} = \underline{k}$

$$\underline{j} \times \underline{k} = \underline{i}$$

$$\underline{k} \times \underline{i} = \underline{j}$$

Right hand rule

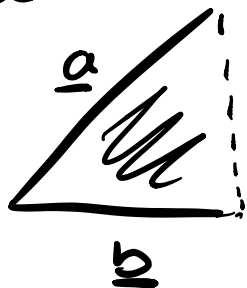


Q6. Area = $|\underline{a} \times \underline{b}|$

$$\underline{a} \times \underline{b} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \times 2 - 1 \times 0 \\ 1 \times 1 - 5 \times 2 \\ 5 \times 0 - 3 \times 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -9 \\ -3 \end{pmatrix}$$

$$\text{so Area} = \left| \begin{pmatrix} 6 \\ 9 \\ -3 \end{pmatrix} \right| = 3 \sqrt{2^2 + 3^2 + 1^2} = 3 \sqrt{14}$$

Q7. Find



then $\frac{1}{2} |a \times b|$

$$\text{so } \underline{a} = \begin{pmatrix} 6 \\ -2 \\ 10 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 8 \end{pmatrix}$$

$$\underline{b} = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \\ -1 \end{pmatrix}$$

$$\text{so } \underline{a} \times \underline{b} = \begin{pmatrix} 1 \\ -5 \\ 8 \end{pmatrix} \times \begin{pmatrix} 0 \\ -5 \\ -1 \end{pmatrix} = \begin{pmatrix} -5 \times -1 - 8 \times -5 \\ 8 \times 0 - 1 \times -1 \\ 1 \times -5 - -5 \times 0 \end{pmatrix} = \begin{pmatrix} 45 \\ 1 \\ -5 \end{pmatrix}$$

$$\text{so Area} = \frac{1}{2} \left| \begin{pmatrix} 45 \\ 1 \\ -5 \end{pmatrix} \right| = \frac{1}{2} \sqrt{45^2 + 1^2 + 5^2} = \frac{\sqrt{2051}}{2}$$

Q8. Two points $\lambda=0$, $\lambda=1$ gives

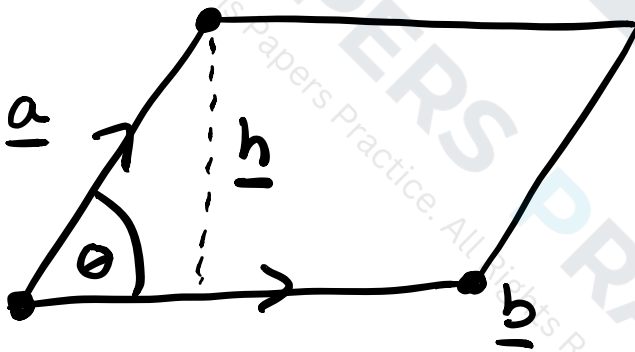
$$\underline{a} = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \quad \underline{b} = \begin{pmatrix} 7 \\ -2 \\ 3 \end{pmatrix}$$

$$\text{Then } \left(\underline{r} - \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix} \right) \times \begin{pmatrix} 7 \\ -2 \\ 3 \end{pmatrix} = \underline{0}$$

Advanced skills

Q 9. A vector perpendicular to $\underline{a}$, $\underline{b}$ in direction by right-handed system with magnitude equal to the area of the parallelogram formed by $\underline{a}$, $\underline{b}$

Q 10.



$$\text{Area} = |\underline{h}| |\underline{b}|$$

$$\text{now } h = |\underline{a}| \sin \theta \hat{n}$$

$$\begin{aligned} \text{SO Area} &= |\underline{a}| \sin \theta \hat{n} |\underline{b}| \\ &= |\underline{a}| |\underline{b}| (\sin \theta \hat{n}) \\ &= |\underline{a} \times \underline{b}| \end{aligned}$$

Q11.

know

$$\underline{i} \times \underline{j} = \underline{k}, \underline{j} \times \underline{k} = \underline{i}$$

$$\underline{k} \times \underline{i} = \underline{j} \text{ and}$$

anti-commutative

$$\text{with } \underline{x} \times \underline{x} = \underline{0}$$

$$\underline{a} = a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k}$$

$$\underline{b} = b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}$$

$$\underline{a} \times \underline{b} = (a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k}) (b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k})$$

$$\begin{aligned} &= a_1 b_1 (\underline{i} \times \underline{i}) + a_1 b_2 (\underline{i} \times \underline{j}) + a_1 b_3 (\underline{i} \times \underline{k}) \\ &\quad + a_2 b_1 (\underline{j} \times \underline{i}) + a_2 b_2 (\underline{j} \times \underline{j}) + a_2 b_3 (\underline{j} \times \underline{k}) \\ &\quad + a_3 b_1 (\underline{k} \times \underline{i}) + a_3 b_2 (\underline{k} \times \underline{j}) + a_3 b_3 (\underline{k} \times \underline{k}) \end{aligned}$$

$$\begin{aligned} &= \underline{0} + a_1 b_2 \underline{k} + a_1 b_3 (-\underline{j}) \\ &\quad + a_2 b_1 (-\underline{k}) + \underline{0} + a_2 b_3 \underline{i} \\ &\quad + a_3 b_1 \underline{j} + a_3 b_2 (-\underline{i}) + \underline{0} \end{aligned}$$

$$\begin{aligned} &= (a_2 b_3 - a_3 b_2) \underline{i} + (a_3 b_1 - a_1 b_3) \underline{j} \\ &\quad + (a_1 b_2 - a_2 b_1) \underline{k} \end{aligned}$$

$$= \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

Q12. AB is

CD is

$$\vec{r}_1 = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ -10 \end{pmatrix} \quad \vec{r}_2 = \begin{pmatrix} k \\ s \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -k \\ k-5 \\ 10 \end{pmatrix}$$

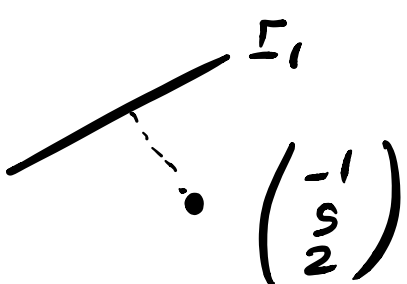
For parallel

$$\begin{pmatrix} 1 \\ -4 \\ -10 \end{pmatrix} = \alpha \begin{pmatrix} -k \\ k-5 \\ 10 \end{pmatrix}$$

$k = -1$ then
 α can be anything

so $\vec{r}_2 = \begin{pmatrix} -1 \\ s \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -4 \\ -10 \end{pmatrix}$ ← any parallel direction.

Take point $\begin{pmatrix} -1 \\ s \\ 2 \end{pmatrix}$ to $\vec{r}_1$



$$\left(\vec{r} - \begin{pmatrix} -1 \\ s \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ -4 \\ -10 \end{pmatrix} = 0 \quad \text{to get point}$$

$$\begin{pmatrix} 3 + \lambda \\ -2 - 4\lambda \\ 3 - 10\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -4 \\ -10 \end{pmatrix} = 0 \Rightarrow \begin{aligned} 3 + \lambda + 8 + 16\lambda - 30 + 100\lambda &= 0 \\ 117\lambda &= 19 \quad \lambda = \frac{19}{117} \end{aligned}$$

so $\begin{pmatrix} 3 + \frac{19}{117} \\ -2 - 4 \times \frac{19}{117} \\ 3 - 10 \times \frac{19}{117} \end{pmatrix} = \frac{1}{117} \begin{pmatrix} 370 \\ -310 \\ 161 \end{pmatrix}$ so distance is $\frac{1}{117} \sqrt{370^2 + 310^2 + 161^2} \approx 4.35$